# Essays

|!|BCEEB0170000001|!| ★ **Cross topic**

Compare and contrast the features of the hormonal coordination in humans and plants.

(10 marks)

##

**Similarities** (Max. 3)

- In both humans and plants, hormones are produced by specific cells in the organism. (1)
- Both human and plants hormones are transported from the site of production to different parts of the organism. (1)
- Both human and plants hormones act on specific cells / organs. (1)

**Differences** (Max. 4)

| Humans | Plants |
|---|---|
| • Coordination is elicited by internal stimuli. | • Coordination is elicited by external stimuli. |
| • Hormones are produced in endocrine glands. | • Hormones are produced in a variety of tissues, usually at meristems / dividing cells. |
| • Hormones are transported more rapidly to the target cells. | • Hormones are transported more slowly to the target cells. |
| • Hormones are transported in the bloodstream. | • Hormones are transported in vascular tissues / from cells to cells. |
| • Responses to hormones are usually faster. | • Responses to hormones are usually slower. |
| • Coordination involves feedback mechanisms. | • There are no feedback mechanisms. |

**Communication** (Max. 3)

(10 marks)

##

| Skills assessed | Marks |
|---|---|
| Understanding basic concepts | 7 |
| Communication | 3 |

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|!|BCEEB0170000002|!| ★

The nervous system and the endocrine system both coordinate body activities by transmitting messages to effectors to produce responses. Using examples, show how nervous coordination and hormonal coordination differ from one another. (11 marks)

##

**Differences** (Max. 8)

| Nervous coordination | Hormonal coordination |
|---|---|
| • Involves nerve impulses (1) | • Involves hormones (1) |
| • Nerve impulses are transmitted by neurones (1) | • Hormones are transported by blood (1) |
| • Response is quick<br>e.g. when a hand touches a sharp object, there is a immediate withdrawal of the hand (1) | • Response is slow<br>e.g. testosterone controls the development of male sex organs and secondary sexual characteristics which takes a long period of time (1) |
| • Response is short-term<br>e.g. the withdrawal reflex stops once the object is removed (1) | • Response may be short-term or long-term<br>e.g. the effects of testosterone last forever (1) |
| • Usually localized<br>e.g. effect usually take places at only one set of muscles (1) | • May affect more than one target organ<br>e.g. insulin stimulates the liver to convert glucose to glycogen, and stimulate body cells to take up more glucose from blood (1) |

**Communication** (Max. 3)

(11 marks)

##

| Skills assessed | Marks |
|---|---|
| Understanding basic concepts | 8 |
| Communication | 3 |

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